Light

GCSE Computer Science – Computational Thinking (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Computational Thinking

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A team of programmers is developing a video game. How does decomposition help them work effectively?

- ○ **A.** It forces all programmers to work on the same code at the same time
- ○ **B.** It eliminates the need for any planning before coding begins
- ○ **C.** It automatically generates all the code needed for the game
- ○ **D.** It breaks the game into modules like graphics, sound, physics, and user input so different programmers can work on different parts independently

Q2 (1 mark)

A flight simulator game models aircraft with speed, altitude, and fuel level, but does not simulate individual air molecules or every rivet on the plane. This is an example of:

- ○ **A.** Abstraction - unnecessary details are removed to focus on essential features that affect gameplay
- ○ **B.** Decomposition - the plane is being broken into smaller parts
- ○ **C.** Pattern recognition - similar planes are being identified
- ○ **D.** An error in the game's design that should be fixed

Q3 (1 mark)

What do arrows (flow lines) represent in a flowchart?

- ○ **A.** The data stored in variables
- ○ **B.** The names of functions in the program
- ○ **C.** Mathematical calculations being performed
- ○ **D.** The direction of flow and sequence in which steps are executed

Q4 (1 mark)

A chef notices that many of their recipes require the same basic technique: sautéing onions and garlic as a base. This recognition is an example of:

- ○ **A.** Abstraction
- ○ **B.** Decomposition
- ○ **C.** Algorithmic thinking

○ **D.** Pattern recognition - identifying a common element that appears across multiple recipes

Q5 (1 mark)

In pseudocode, what does the following line do?

```
score = score + 10
```

○ **A.** Checks if score equals score plus 10

○ **B.** Creates a new variable called 'score + 10'

○ **C.** Takes the current value of score, adds 10 to it, and stores the result back in score

○ **D.** Displays the value of score on the screen

Q6 (1 mark)

In a supermarket self-checkout system, which of the following is an input?

○ **A.** The barcode scanned from a product

○ **B.** The receipt printed at the end

○ **C.** The total price displayed on screen

○ **D.** The bag area weight warning message

Q7 (1 mark)

Which of the following everyday activities is best described as following an algorithm?

- ○ **A.** Following a recipe to bake a cake with specific steps in order
- ○ **B.** Daydreaming about a holiday destination
- ○ **C.** Randomly opening kitchen cupboards looking for a snack
- ○ **D.** Deciding what to watch on television based on mood

Q8 (1 mark)

In a structure diagram, what does it mean when one box is positioned directly below and connected to another box?

- ○ **A.** The upper task should be deleted and replaced with the lower one
- ○ **B.** The two tasks happen at the same time
- ○ **C.** The lower box is a sub-task or component of the task in the upper box
- ○ **D.** The tasks are completely unrelated to each other

Q9 (1 mark)

In pseudocode, what is the difference between = and ==?

- ○ **A.** = assigns a value to a variable; == compares two values to check if they are equal
- ○ **B.** There is no difference - they mean the same thing
- ○ **C.** = is used for numbers; == is used for text

- ○ **D.** = is old syntax; == is new syntax

Q10 (1 mark)

What is computational thinking?

- ○ **A.** A problem-solving approach that uses techniques like decomposition, abstraction, pattern recognition, and algorithmic thinking
- ○ **B.** Thinking about computers and how they are manufactured
- ○ **C.** The ability to type quickly on a keyboard
- ○ **D.** Memorising programming language syntax

Q11 (1 mark)

In a flowchart for calculating the area of a rectangle, which symbol would contain 'area = length × width'?

- ○ **A.** Diamond - because it's making a decision
- ○ **B.** Parallelogram - because it's an input
- ○ **C.** Rectangle - because it's a process or calculation
- ○ **D.** Oval - because it's the end of the algorithm

Q12 (1 mark)

Why is abstraction important when solving complex problems?

- ○ **A.** It makes problems more complicated so they seem more impressive
- ○ **B.** It eliminates the need for any actual solution
- ○ **C.** It converts all problems into mathematical equations
- ○ **D.** It simplifies problems by hiding unnecessary complexity, allowing focus on what's essential

Q13 (2 marks)

A student is asked to create a school website. They decompose the project into: Home page, News section, Staff directory, Contact form, Photo gallery.
Why is this a good example of decomposition?

- ○ **A.** Because it lists five things, and five is a good number
- ○ **B.** Because each component is a distinct functional area that can be designed, built, and tested independently while together forming the complete website
- ○ **C.** Because all websites must have exactly these five sections
- ○ **D.** Because the components are listed in alphabetical order

Q14 (2 marks)

Consider the following pseudocode:

```
name = "COMPUTER"
result = name.substring(0, 3)
OUTPUT result
```

What will be output?

- ○ **A.** COMPUTER
- ○ **B.** COM
- ○ **C.** OMPUTER
- ○ **D.** C

Q15 (2 marks)

A flowchart shows: Start → Set total to 0 → Set counter to 1 → Is counter > 5? → (No) INPUT number → total = total + number → counter = counter + 1 → [back to decision] → (Yes) OUTPUT total → End
What does this algorithm do?

- ○ **A.** Outputs the number 5
- ○ **B.** Counts from 1 to 5
- ○ **C.** Inputs five numbers and outputs their sum
- ○ **D.** Multiplies five numbers together

Q16 (2 marks)

A programmer writes code to convert temperatures from Celsius to Fahrenheit, then separately writes code to convert temperatures from Celsius to Kelvin, then writes code to convert distances from miles to kilometres.
What pattern should they recognise?

- ○ **A.** All three involve temperature, so they should be combined
- ○ **B.** All three involve conversion using a formula, so a generalised 'convert' function accepting values and conversion factors could replace all three

- ○ **C.** The code should be deleted as conversions are not useful
- ○ **D.** No pattern exists because they convert different things

Q17 (2 marks)

An automatic car wash system works as follows: the customer selects a wash type, pays with a card, then drives into the wash bay where sensors detect the car and machines apply water, soap, and brushes before drying.
Identify the processes in this system.

- ○ **A.** The processes are: wash type selection and card payment only
- ○ **B.** The processes are: detecting car position, applying water, applying soap, operating brushes, and drying - the actual transformative operations performed on the car
- ○ **C.** The processes are: the clean car and the receipt only
- ○ **D.** There are no processes in a car wash system

Q18 (2 marks)

Consider the following pseudocode:

```
array colours = ["Red", "Green", "Blue", "Yellow"]
OUTPUT colours[2]
```

What will be output?

- ○ **A.** Red
- ○ **B.** Green
- ○ **C.** Yellow

- ○ **D.** Blue

Q19 (2 marks)

A flowchart has a diamond (decision) shape with 'Is age >= 18?' inside it. The 'Yes' path leads to another diamond with 'Has valid ID?' inside it. This structure represents:

- ○ **A.** A single decision with two outcomes
- ○ **B.** Nested selection where a second condition is only checked if the first condition is true
- ○ **C.** A loop that repeats twice
- ○ **D.** An error in the flowchart design

Q20 (2 marks)

A programmer needs to validate that a user-entered UK postcode has between 5 and 8 characters. Which algorithm structure is most appropriate?

- ○ **A.** Use a loop to repeatedly ask for the postcode until it's exactly 7 characters
- ○ **B.** Use selection to check if the length is >= 5 AND <= 8, accepting if true, rejecting if false
- ○ **C.** Use no validation as users always enter correct data
- ○ **D.** Use a loop that counts from 5 to 8

Q21 (2 marks)

A structure diagram shows 'Email System' at the top, with 'Compose Email', 'Inbox', 'Sent Items', and 'Settings' at the second level. Under 'Compose Email' are 'Add Recipients', 'Write Message', and 'Attach Files'. What can we conclude about 'Add Recipients'?

- ○ **A.** It is the most important part of the entire email system
- ○ **B.** It is the parent of 'Compose Email'
- ○ **C.** It is at the same level as 'Inbox' and 'Sent Items'
- ○ **D.** It is a sub-component of 'Compose Email', which is itself a component of the 'Email System'

Q22 (2 marks)

Consider the following pseudocode:

```
IF temperature > 30 OR raining == true THEN
    OUTPUT "Stay indoors"
ELSE
    OUTPUT "Go outside"
ENDIF
```

When will "Stay indoors" be output?

- ○ **A.** Only when it is both hot (>30) AND raining at the same time
- ○ **B.** When the temperature is above 30 OR when it is raining, or both - either condition alone is sufficient
- ○ **C.** Only when the temperature is exactly 30 degrees
- ○ **D.** Only when it is not raining

Q23 (2 marks)

A city creates a bus route map showing only bus stops and the routes connecting them, without showing every building, tree, or road curve. What does this demonstrate about abstraction?

- ○ **A.** Abstraction keeps only information relevant to the purpose - bus travellers need stops and routes, not building details
- ○ **B.** Abstraction removes all information, leaving a blank page
- ○ **C.** Abstraction adds extra details to make maps more interesting
- ○ **D.** Abstraction is only used in computer science, not in everyday life

Q24 (3 marks)

A software company needs to build mobile apps for three different clients: a restaurant booking app, a hotel booking app, and a cinema booking app.
How would decomposition and pattern recognition together improve efficiency?

- ○ **A.** Neither technique is useful for mobile app development
- ○ **B.** Decomposition would break each app into separate isolated projects with no shared components
- ○ **C.** Decomposition identifies common components across all apps (user authentication, booking calendar, payment processing, notifications); pattern recognition reveals these are reusable. The company builds these once as shared modules, then customises unique elements (menu browsing, room selection, seat choosing) for each client, dramatically reducing development time.
- ○ **D.** Pattern recognition means copying the restaurant app exactly for hotel and cinema use

Q25 (3 marks)

Consider the following pseudocode:

```
array nums = [5, 2, 8, 1, 9]
min = nums[0]
minIndex = 0
FOR i = 1 TO 4
    IF nums[i] < min THEN
        min = nums[i]
        minIndex = i
    ENDIF
ENDFOR
OUTPUT min
OUTPUT minIndex
```

Trace through this algorithm and determine what will be output and what the algorithm's purpose is.

- ○ **A.** Output: min = 9, minIndex = 4; purpose is to find the maximum value
- ○ **B.** Output: min = 1, minIndex = 3; purpose is to find the minimum value and its position in the array
- ○ **C.** Output: min = 5, minIndex = 0; purpose is to output the first element
- ○ **D.** Output: min = 2, minIndex = 1; purpose is to find the second element

Q26 (3 marks)

A programmer needs to create a flowchart for a login system with the following requirements: accept username and password; check if username exists; if yes, check if password matches; allow 3 attempts before locking the account.
Analyse which constructs would be needed.

- ○ **A.** Only sequence is needed - the steps simply follow one after another

○ **B.** Only selection is needed - there are just two IF statements to check username and password

○ **C.** The flowchart needs sequence (ordered steps), selection (checking username exists, checking password matches), and iteration (allowing up to 3 attempts). An outer loop controls attempts; nested selections check credentials; the loop exits on success or after 3 failures.

○ **D.** Flowcharts cannot represent login systems

Q27 (3 marks)

A company is building a GPS navigation system. The engineering team debates what level of abstraction to use for modelling roads. Analyse the trade-offs between different abstraction levels.

○ **A.** Always use the lowest abstraction with maximum detail for best results

○ **B.** Always use the highest abstraction with minimum detail to save time

○ **C.** Low abstraction (individual lane markings, traffic light phases, road surface conditions) provides precision but requires massive data storage and processing power, making real-time navigation impractical. High abstraction (straight-line distances between cities) is fast but ignores actual roads, giving useless directions. Medium abstraction (road segments with average speeds, distances, junction connections) balances accuracy with computational feasibility for practical real-time navigation.

○ **D.** Abstraction level doesn't affect GPS system performance

Q28 (3 marks)

A programmer notices their code contains these three similar loops:

- Loop 1: Count how many students scored above 50
- Loop 2: Count how many products cost less than £10

- Loop 3: Count how many temperatures are below freezing
  Apply pattern recognition to design an optimised solution.

○ **A.** The loops are completely different and cannot be optimised

○ **B.** Delete all three loops as counting is unnecessary

○ **C.** Combine all three loops into one that counts students, products, and temperatures simultaneously

○ **D.** The pattern is: iterate through a list, apply a condition to each item, count how many meet the condition. Create a generalised function 'countMatching(list, condition)' that accepts any list and any condition, returning the count of items satisfying that condition. This single function replaces all three loops and handles any future similar requirements.

Q29 (3 marks)

A hospital wants to reduce patient waiting times in the emergency department. Apply computational thinking to analyse how this problem could be approached.

○ **A.** Computational thinking cannot be applied to healthcare problems

○ **B.** Simply hire more doctors without any analysis

○ **C.** Decomposition breaks the problem into: patient arrival patterns, triage process, treatment procedures, staff scheduling, bed availability, and discharge processes. Abstraction models patients by severity category rather than individual medical details. Pattern recognition identifies peak times and common bottlenecks. Algorithmic thinking designs optimised scheduling and patient flow procedures. Together, these provide a systematic framework for understanding and improving the complex system.

○ **D.** Pattern recognition alone is sufficient to solve this problem

Q30 (3 marks)

Two programmers write algorithms to find a specific book in a library catalogue of 10,000 books sorted alphabetically by title.
Programmer A: Start from the first book and check each one sequentially until found.
Programmer B: Check the middle book; if target comes before alphabetically, search the first half; if after, search the second half; repeat on the chosen half.
Compare the efficiency and explain why one approach is superior for this scenario.

- ○ **A.** Programmer A's approach is more efficient because it's simpler to understand
- ○ **B.** Both approaches will always check the same number of books
- ○ **C.** Programmer A uses linear search (up to 10,000 comparisons worst case). Programmer B uses binary search (about 14 comparisons worst case, since $\log_2 10000 \approx 13.3$). For sorted data, binary search is dramatically more efficient - each comparison eliminates half the remaining possibilities. The trade-off is that binary search requires sorted data and is more complex to implement correctly.
- ○ **D.** Programmer B's approach won't work on sorted data

Total: **55** marks • Time: **30 min**

↑ Back to top

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